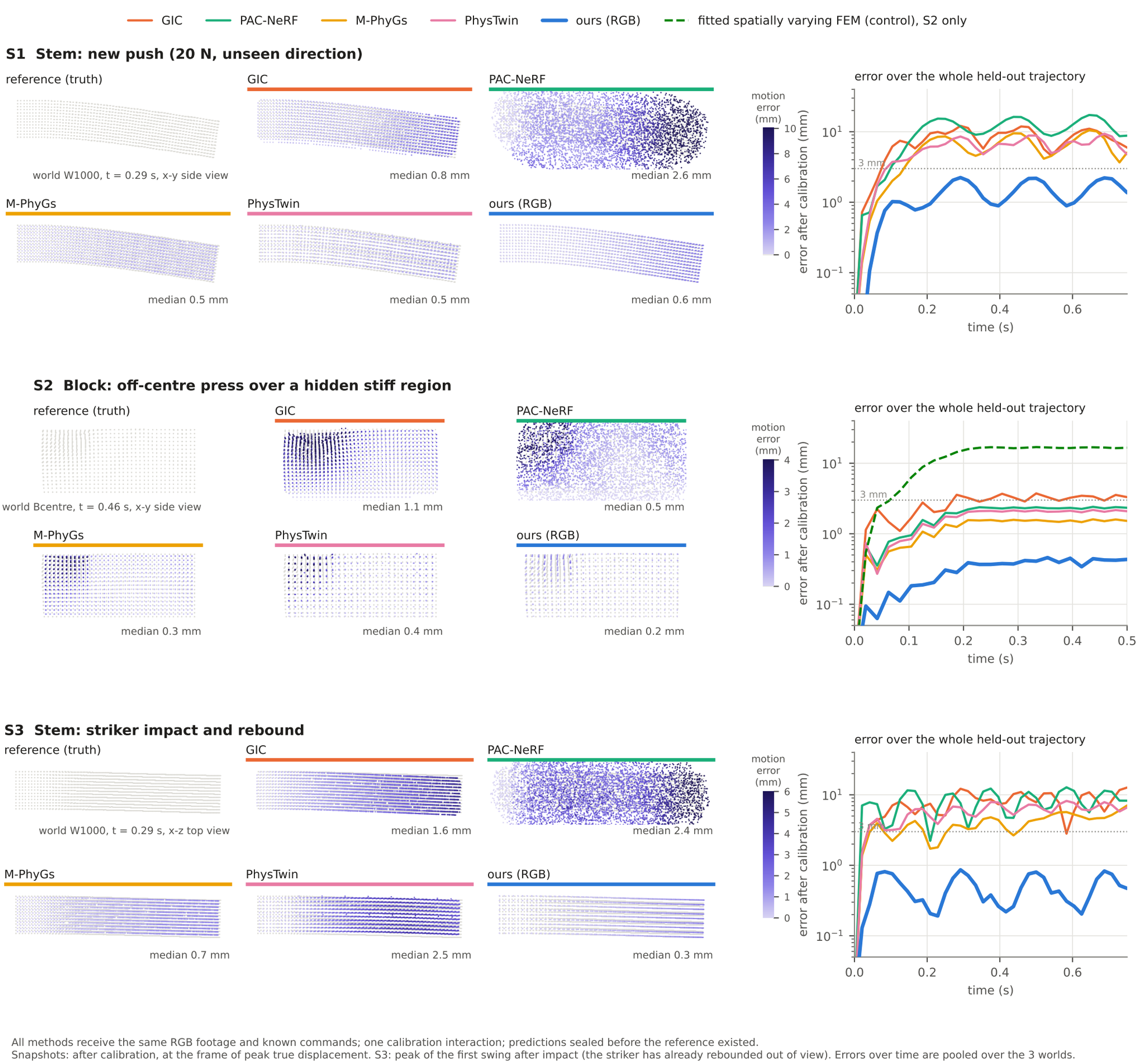


All methods receive the same RGB footage and known commands; one calibration interaction; predictions sealed before the reference existed.
Snapshots: after calibration, at the frame of peak true displacement. S3: peak of the first swing after impact (the striker has already rebounded out of view). Errors over time are pooled over the 3 worlds.